

PROPERTIES OF MATTER – I (04)

Aims

- A. To determine the surface tension of a soap solution using the Ferguson's method.
- B. To determine the viscosity of air.

Materials and Apparatus

Convex lens, ground glass, light bulb, white screen, pressure funnel, beaker, manometer, soap solution, capillary tubes, meter ruler, stop watch, cathetometer, lens holder, stands, rubber tubes and lab jack.

Theory

Part A:

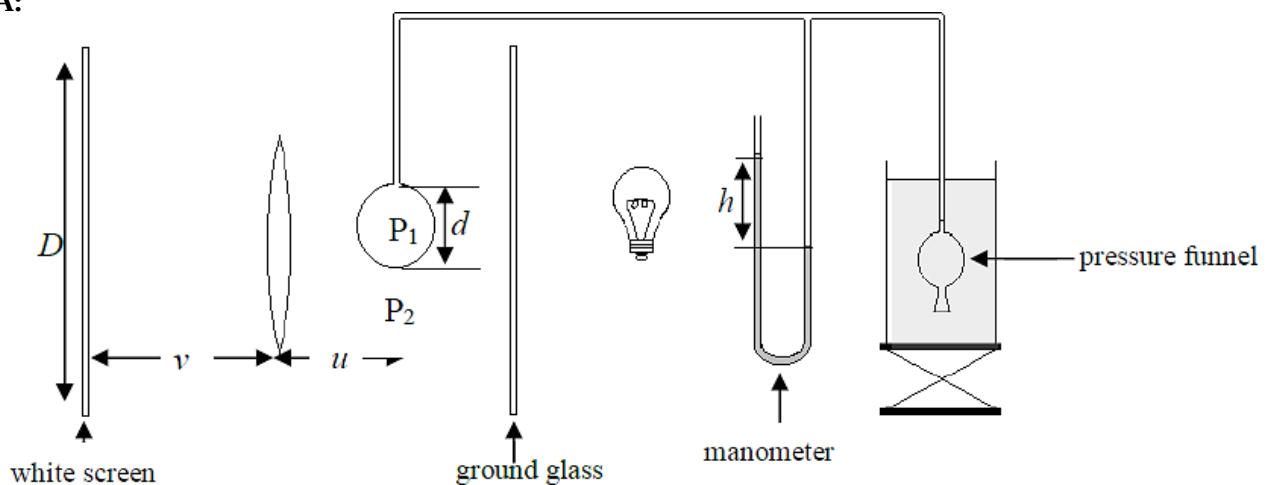


Fig. 1. The experimental setup of the Ferguson's method

Where, d : Diameter of the soap bubble, D : Diameter of the image, u : Objective distance, v : Image distance, P_1 : Pressure inside the soap bubble, P_2 : Pressure outside the soap bubble, h : height difference between the liquid levels in manometer

The excess pressure inside a soap bubble of radius R is given by;

$$\Delta P = \frac{4T}{R} = (P_1 - P_2) \quad (1)$$

where,

T - Surface tension of the soap solution.

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The excess pressure inside the bubble can be measured from the manometer as,

$$\Delta P = h\rho g \quad (2)$$

where,

h : Height difference of the liquid columns in the manometer,

ρ : Density of the manometer liquid,

g : Acceleration due to gravity.

From equations (1) and (2),

$$h = \left(\frac{4T}{\rho g}\right) \frac{1}{R} \quad (3)$$

T can be found from the gradient of the graph of h vs $\frac{1}{R}$

A lens arrangement can be used as shown in Fig.1 to find the diameter d of the soap bubble using the equation of magnification,

$$\frac{D}{d} = \frac{v}{u} \quad (4)$$

Part B:

The rate of flow of a fluid through a narrow tube is given by Poiseuille's equation;

$$\frac{dQ}{dt} = \frac{\pi P r^4}{8\eta l} \quad (5)$$

where,

r : Radius of the capillary tube

P : Pressure difference at the two ends of the capillary tube

η : Coefficient of viscosity of the fluid

l : Length of the capillary tube.

$$\frac{dQ}{dt} = - \frac{dv}{dt} \quad (6)$$

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where,

v - Volume of the soap bubble $v = \frac{4}{3}\pi R^3$ and

$$\frac{dv}{dt} = 4\pi R^2 \frac{dR}{dt} \quad (7)$$

Also,

$$P_1 - P_2 = \frac{4T}{R} \quad (8)$$

$$R^3 dR = - \frac{Tr^4}{8\eta l} dt \quad (9)$$

$$\int_{R_0}^R R^3 dR = - \frac{Tr^4}{8\eta l} \int_0^t dt \quad (10)$$

$$R^4 = - \frac{Tr^4}{2\eta l} t + R_0^4 \quad (11)$$

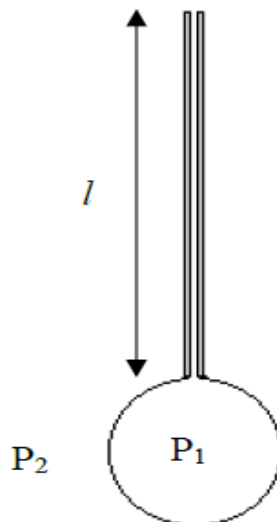


Fig. 2. The experimental setup

Where, P_1 : Internal pressure of the soap bubble,

P_2 : Atmospheric pressure, l : Length of the capillary tube

η can be determined from the gradient of the graph R^4 vs t .

Procedure

Part A:

- a) Set the apparatus as shown in **Fig. 1**.
- b) Adjust the screen and the lens to get a sharp image of the capillary tube on the screen.

Q: What is the purpose of using a ground glass?

Q: What are the measurements required to be taken at this step?
- c) Gently dip the tip of the capillary tube in the soap solution once.
- d) Create a soap bubble by gently increasing the height of the lab jack.
- e) Measure the horizontal and vertical diameters of the image of the soap bubble on the screen.

Q: How do you determine the real diameter of the soap bubble?
- f) Measure the corresponding pressure difference in the manometer using the cathetometer.

Q: What are the important properties that the manometer liquid should have?
- g) Repeat above two steps for different D values while adjusting the height of the lab jack.
- h) Note down the room temperature.

Q: Why is it necessary to record the temperature?
- i) Find the surface tension of the soap solution by plotting a graph.

Part B:

- a) Blow up a soap bubble in the lower end of the capillary tube and allow air to flow out to the atmosphere through the capillary tube.

Q: How do you blow up the air bubble?
- b) Measure the diameter of the bubble with time.

Q: How do you obtain the measurements?
- c) Plot the graph R^4 vs t .
- d) Determine the viscosity of air inside the bubble using the graph.

Q: What are the other readings require?